

Two Margins of International Track-and-Field Recruitment to NCAA Division I: Wealth, Population, and the Limits of Macro Explanation

Alina Malkova
Florida Institute of Technology

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Abstract

We construct a country-level panel of international student-athletes in U.S. NCAA Division I rosters for the 2024–25 / 2025–26 season. Athlete-level records are scraped from athletic-department websites for soccer ($n = 660$) and track and field / cross country ($n = 18,657$); 96% of the international sample competes in track, so we characterise the empirical setting as the international track-and-field recruitment market. We decompose athlete supply along two margins. **On the extensive margin** (whether a country sends any athletes at all), log GDP per capita and log population are strong independent predictors ($\hat{\beta} = 0.52$ and 0.18 , both $p < 0.01$ in a probit on 191 countries); political stability is not. **On the intensive margin** (among 102 sender countries), the elasticity of athletes with respect to the 15–24-year-old cohort is approximately 0.33 —strongly sublinear—and log GDP per capita is the only additional robust predictor ($\hat{\beta} = 0.37$, $p < 0.05$). Political stability, disaster events, conflict years, refugee outflows, and macroeconomic shocks are statistically indistinguishable from zero once we estimate the population elasticity freely instead of imposing the unit-elasticity restriction implicit in athletes-per-cohort rate models. We document this in Section 3.3: the rate-model coefficient on political stability is $+1.12$ ($p < 0.01$) and on disaster events is -0.74 ($p < 0.01$), but both collapse once log pop enters the equation as a free regressor. We thus retract earlier framings of this work that emphasised institutional quality as an intensive-margin driver. The honest finding is smaller: wealth and population scale govern entry into the international NCAA pipeline; conditional on entry, wealth governs intensity with diminishing returns to size, and the macro covariates we have do not explain the rest. R^2 in the count model is 0.20 – 0.25 , with the residual most likely attributable to country-specific recruitment pipelines (Kenya–Iten, Jamaica–MVP, the Caribbean sprint network) that macro variables cannot proxy.

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). Only 68 of 1,830 international athletes compete in soccer, so we report the combined sample in headline tables but characterise the empirical setting throughout as the *international track-and-field recruitment market*; the soccer panel is too small to identify country-level structural effects on its own.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms (“U.K.”, “B.C.”, “Czech Republic”). The mapping resolves 87.6% of track and 99.8% of soccer hometowns. Of 19,317 total athletes, **1,830 are international** (country code other than USA), and 1,819 of those merge cleanly to the 2023 World Bank macro panel.

Country covariates are taken from World Bank WDI (GDP per capita PPP, population, 15–24-year-old cohort 2018–2022 average, unemployment, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO armed-conflict data, EM-DAT natural disasters (event counts and total deaths), the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics (refugees and asylum seekers by country of origin). For each country we use both 2023 cross-sections and 2010–2023 cumulative aggregates. The country panel used for intensive-margin analysis contains the **102 countries with at least one international athlete**. The extensive-margin probit uses a broader universe of 214 countries with non-missing macro covariates, of which 107 send at least one athlete.

An initial project plan proposed a principal-component-based “Economic Distance Index.” We ultimately preferred reporting the individual covariates because their separate identification (GDP vs. political stability vs. disasters vs. population scale) is the substantive question, and—as we show in Section 3.3—imposing even seemingly innocuous restrictions on the joint covariate structure (such as the unit population elasticity implicit in a per-cohort rate outcome) can produce coefficients that look significant but vanish under freer specifications.

Table 1. Descriptive statistics by income tier

Countries are split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. sta
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

The descriptive table foreshadows the two-margin structure we develop below. Q1 supplies many athletes ($n = 533$) from large populations (the well-known Kenya / Nigeria / Jamaica pipelines) but at a very low per-cohort rate. Q4 supplies more athletes ($n = 825$) from much smaller populations, at a per-cohort rate sixteen times higher than Q1. The cohort-rate gradient looks dramatic but is heavily driven by the population denominator: the median Q1 country has 4.1 million 15–24-year-olds vs. 0.9 million in Q4. We return to this in Section 3.3, where we show that the rate gradient attenuates substantially once we allow athletes to scale sub-proportionally with population.

2 Specification

We decompose international NCAA athlete supply into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c), \quad (1)$$

estimated by probit on the 191 countries with non-missing covariates in the full universe.

The intensive margin is the per-country athlete count among the 102 senders. We deliberately use the count outcome with $\log \text{pop}_{15-24,c}$ as a free regressor,

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{pop}_{15-24,c} + \beta_2 \log \text{GDPpc}_c + \beta_3 \text{polstab}_c + \mathbf{x}'_c \boldsymbol{\delta} + \varepsilon_c, \quad (2)$$

rather than the per-cohort rate outcome used in the previous draft of this work. Section 3.3 shows that imposing the unit-elasticity restriction $\beta_1 = -1$ (the rate-model assumption) is rejected sharply by the data: the estimated $\hat{\beta}_1$ in the count model is approximately $+0.33$, meaning athletes scale roughly as $\text{pop}^{0.33}$. Imposing -1 in the rate model forces this mis-specification into other coefficients, generating spurious “effects” for political stability and disaster events. The free-elasticity count model is the honest test.

We report Huber–White (HC1) heteroskedasticity-robust standard errors throughout.

3 Results

3.1 Extensive margin

Probit: Pr(sends any athletes)	
log GDP per capita PPP	+0.52*** (0.11)
log population	+0.18*** (0.06)
Political stability (WGI)	+0.19 (0.17)
n	191
Pseudo R^2	0.19

Income and size are strong, independent predictors of whether a country participates in the international NCAA recruitment market at all. Political stability is statistically zero on this margin. The average marginal effect of $\log \text{GDPpc}$ on $\Pr(\text{sends})$ is approximately $+0.20$ per log-unit at the sample mean: a country one log-unit richer ($\approx 2.7\times$) is 20 percentage points more likely to send at least one athlete to NCAA D-I, controlling for population and stability.

3.2 Intensive margin

Among the 102 sender countries, the count model in Equation ?? estimates the country-level log–log relationship between athletes and the macro covariates.

Coefficient	(1)	(2)	(3)	(4) drop top-20
log pop _{15–24}	+ 0.35 ^{***} (0.07)	+ 0.33 ^{***} (0.11)	+0.32 ^{***} (0.11)	+0.16 ^{**} (0.08)
log GDPpc	+ 0.37 ^{**} (0.16)	+ 0.37 ^{**} (0.16)	+0.26 (0.19)	+ 0.37 ^{***} (0.11)
Political stability	+0.38 (0.24)	+0.37 (0.25)	+0.23 (0.31)	−0.11 (0.18)
log(1 + disaster events 2010–23)	—	+0.03 (0.19)	+0.10 (0.19)	−0.05 (0.12)
Econ. shocks 2010–23	—	—	+0.07 (0.09)	—
Years with conflict 2010–23	—	—	+0.02 (0.05)	—
log(1 + refugees 2010–23)	—	—	−0.13 (0.08)	—
n	100	100	100	80
R^2	0.21	0.21	0.24	0.15

HC1 robust SEs in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Three things to flag.

Athletes scale sublinearly with cohort size. The estimated elasticity is $\hat{\beta}_1 \approx 0.33$ in the base specification and is stable across columns. A doubling of the 15–24 cohort is associated with $2^{0.33} \approx 1.26\times$ more athletes, not $2\times$. The most plausible mechanism is that the US recruitment system has finite capacity per country: the marginal NCAA program does not scale its international intake one-for-one with the size of any single source country. Small wealthy countries (Norway, Iceland, the Bahamas, Trinidad) are therefore systematically over-represented per capita relative to populous ones (India, Indonesia, Pakistan). This is the dominant intensive-margin fact in the data.

Log GDP per capita is the only other robust predictor. The coefficient is $\hat{\beta}_2 \approx 0.37$ ($p < 0.05$) in columns (1), (2), and (4). It attenuates to +0.26 (n.s.) in column (3) when all four shock variables are entered jointly, which we attribute to limited statistical power with only 100 observations and seven right-hand-side variables; the drop-top-20 specification in column (4) returns $\hat{\beta}_2$ to +0.37 ($p < 0.01$), supporting the interpretation that the column-(3) attenuation is a degrees-of-freedom artefact rather than substantive.

Macro shocks and institutional quality do not add explanatory power. Disaster events, political stability, economic shocks, conflict years, and refugee outflows are all individually insignificant in columns (2)–(4). Most have point estimates well within one standard error of zero. The combined increase in R^2 from spec (1) to spec (3) is only 0.03. We cannot reject the null that, conditional on GDP per capita and the sublinear population term, these variables have no additional intensive-margin effect.

3.3 Methodological note: the rate-model artefact

An earlier draft of this paper reported intensive-margin results using $\log(\text{athletes}/\text{pop}_{15-24})$ as the outcome and *omitting* $\log \text{pop}$ from the right-hand side. That specification implicitly imposes a unit elasticity of athletes with respect to population:

$$\log(\text{athletes}/\text{pop}) = \log(\text{athletes}) - \log(\text{pop}),$$

so a coefficient of zero on the right-hand side of the rate regression corresponds to $\beta_{\text{pop}} = 1$ in the count regression. The data reject this restriction: the free-elasticity estimate is $\hat{\beta}_{\text{pop}} \approx 0.33$, not 1. Imposing the wrong restriction forces the resulting mis-specification error into the other coefficients. The table below shows the consequence for the variables that previously appeared as the paper’s headline findings.

Coefficient	Rate model (implicit $\beta_{\text{pop}} = 1$)	Count model (β_{pop} free)	Difference
log GDPpc	+0.25 (0.22)	+ 0.37** (0.16)	—
Political stability	+ 1.12*** (0.27)	+0.37 (0.25)	collapses
log(1 + disaster events)	− 0.74*** (0.16)	+0.03 (0.19)	collapses
log pop _{15–24}	(constrained = −1)	+ 0.33*** (0.11)	—
R^2	0.53	0.21	—

The rate model’s apparent “political stability dominates” and “disaster events suppress supply” findings are size-effect artefacts. Small countries tend to be high on the WGI political- stability scale (Norway, Iceland, Switzerland, New Zealand) and tend to have few EM-DAT disaster events (a function of country area and hazard exposure as much as governance), and large countries tend to have the opposite. When we constrain the population elasticity to -1 in the rate model, this small-country / large-country correlation gets loaded onto the stability and disaster coefficients that should have been zero. Freeing the elasticity reveals the underlying null result.

This pitfall is not specific to our paper; it applies to any cross-country regression that uses a per-capita or per-cohort outcome without verifying that the implicit elasticity matches the data. We document it here because it materially changed the empirical conclusions of an earlier version of this work, and because the same mechanism likely affects other applied work in international recruitment and migration that uses per-capita rates as outcomes.

3.4 Sport-specific patterns

The international concentration is in track. The track-only count panel ($n = 99$) is essentially the headline result: log pop_{15–24} +0.32 ($p < 0.01$), log GDPpc +0.37 ($p < 0.05$), political stability +0.36 (n.s.). The soccer panel ($n = 24$) is too small for clean identification: only log pop approaches conventional significance. The paper’s findings should be understood as describing the international track-and-field market specifically; international soccer recruitment likely operates through school-level network channels (specific academies, alumni connections) that a country-year macro design cannot capture, and we leave that question to future work.

3.5 Robustness

We test the count-model intensive-margin result against several alternative specifications.

Disaster deaths instead of events. The previous draft used log(1 + disaster deaths) rather than events. The deaths variable in the rate model gives -0.23 ($p < 0.01$); in the count model with log pop_{15–24} freely estimated it collapses to -0.04 (n.s.). The deaths effect was driven by the population-size confound, not by the catastrophic- event countries (the top-5 disaster-death countries in our sample are Russia, Turkey, Italy, India, and Japan—all large, all substantial NCAA senders—rather than the small high-fatality cases of Haiti, Myanmar, or Nepal that one might naively expect). Events is the cleaner measure even before the size correction, but neither survives the size correction.

Refugees: counts vs. rates. The rate model gave a positive marginal coefficient on refugees per cohort (+0.74, n.s. with disasters in the model). On raw counts with log pop controlled, refugee outflows enter at -0.13 ($p < 0.10$). The sign flip is consistent with the war-disrupted-cohort

denominator story: countries with large refugee outflows (Syria, Afghanistan, Venezuela, South Sudan) have shrinking 15–24 cohorts, so the rate ratio is inflated independent of any underlying recruitment effect. In the count model, conditional on size, refugee outflows do not predict athlete supply.

Drop-top-20 senders. Dropping the 20 countries with the most international athletes (column 4 of Table 3) shrinks the sample to 80 countries and the R^2 to 0.15. The GDP per capita coefficient stays at +0.37 ($p < 0.01$); the population elasticity falls to +0.16 ($p < 0.05$); political stability becomes negative and insignificant. We read this as: the GDP signal is genuine, the population-scaling signal is diluted when the largest senders are removed (a few thousand of the $\sim 18,000$ track athletes are concentrated in those 20 countries), and the political-stability coefficient remains noise.

Bartik shift-share. We constructed a Bartik-style instrument from country shares across nine track-event groups and leave-one-out aggregate event flows. The first-stage F -statistic on the Bartik exposure as an instrument for log GDPpc is 13.5, below the modern threshold of $F > 104.7$ for valid 2SLS t -tests at the 5% level (?), so we do not rely on the IV. As a covariate the Bartik enters at +0.95 ($p < 0.10$) and absorbs the GDP coefficient (a mechanical collinearity result, since sport mix is correlated with country wealth). The Bartik specification is reported in `output/bartik_results.txt` for transparency and is not load-bearing in this paper.

4 Discussion

Three substantive conclusions emerge from the country-level analysis, and a fourth from what we tried but could not find.

First, **wealth and population scale govern entry into the international NCAA pipeline.** Both log GDPpc and log pop are large and statistically significant in the extensive-margin probit; political stability is not. The mechanism is the fixed-cost story familiar from migration economics: producing a recruitable athlete requires some minimum infrastructure (clubs, schools, transport, equipment, parental capacity) and the discovery cost on the US side, both of which scale with national resource flow. Below the corresponding income threshold, no athletes are produced regardless of how well-governed the country is.

Second, **conditional on entry, the intensive margin is dominated by sublinear population scaling and a small wealth gradient.** The cohort-population elasticity of 0.33 is the largest single fact in the country panel: a doubling of source-country 15–24 population is associated with only $\sim 26\%$ more NCAA athletes, not the proportional doubling that a constant-returns supply curve would predict. This is consistent with diminishing-returns recruitment at the country level: as a source country grows, the marginal recruit is harder to find, the marginal coach has less information, and the US side’s bilateral recruiting capacity saturates. The GDP-per-capita coefficient of 0.37 adds a modest wealth gradient on top: among countries of similar size, a richer one produces about $\exp(0.37 \cdot \ln 2) - 1 \approx 29\%$ more athletes per doubling of GDP per capita.

Third, **institutional quality and macro shocks do not have detectable intensive-margin effects** once population scale is freely estimated. Political stability, disaster events, disaster deaths, conflict years, refugee outflows, CPI inflation, and terms-of-trade shocks are all individually insignificant in the count model. A previous draft of this work emphasised political stability as the robust intensive-margin driver; we now show that result was an artefact of imposing a unit population

elasticity. The honest conclusion is that we cannot detect institutional or shock effects beyond the joint role of wealth and scale.

Fourth, and most importantly, **the country-level macro design explains only about a fifth of the variation in athlete supply** ($R^2 \approx 0.21$ in the base count model, 0.24 with the full shock battery). More than three-quarters of the cross-country variation in international NCAA track-and-field representation is not explained by GDP, population, governance, conflict, disasters, shocks, inflation, or refugees. We attribute the bulk of this residual to country-specific recruitment pipelines: the Kenyan-distance-running ecosystem centered on Iten, the MVP and Racers Track Club networks that dominate Jamaican sprinting, similar Caribbean conduits, the Norwegian and Australian middle-distance scenes, the German and Polish throwing pipelines, and so on. These are pipeline structures that macro covariates cannot capture, and a complete model of international NCAA supply would require sport-level network data we do not have. Our reading is that the ceiling on what country-year macro variables can explain in this domain is closer to $R^2 = 0.30$ than 0.60.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26. Macro covariates are 2023 plus 2010–23 aggregates. A descriptive pseudo-panel constructed from athletes’ class-year (Fr/So/Jr/Sr/Gr) gives at most five entry cohorts. A static DiD using country and year fixed effects with COVID-19 stringency interacted with the 2021–22 entry cohorts gives a small positive coefficient (+0.07, $p = 0.06$, country-clustered SEs). We report this in `output/event_study.txt` as a descriptive exercise rather than an identified event study: there is no clean pre-period (the cohort 2020 is itself a COVID-affected cohort), and the class-year-to-entry-year mapping is imputation-based.
- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (school-page rendering issues that systematically affect schools with JS-rendered rosters).
- **Track-and-field dominance.** 96% of the international sample competes in track. The paper’s findings should be understood as describing the international track-and-field recruitment market specifically.
- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals (only 5-year bands as percentages of gender-specific totals). Rescaling the 15–24 cohort by a constant fraction is observationally equivalent in the count model (the constant is absorbed by the intercept), so denominator choice does not change our slope estimates, but a finer-grained cohort variable from national censuses would be valuable.
- **Athletes per capita is not athletic talent per capita.** Recruiting selection, scholarship caps, language and visa frictions, and US schools’ bilateral relationships filter the population. We observe the *combined* effect of supply, selection, and demand.
- **Bartik weak instrument.** The shift-share construction on track-event composition has a first-stage $F = 13.5$, below the modern $F > 104.7$ threshold (?) for valid 2SLS t -tests at the 5% level. We use it only as a transparency check, not as identification.

- **Modest R^2 .** The residual 75–80% of cross-country variation in athlete supply is most plausibly driven by country- specific recruitment pipelines that macro covariates cannot proxy.

6 Reproducibility

All results in this paper are reproducible from the local Python scripts in the project repository:

- `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`: build the roster data.
- `run_country_mapping.py`: resolves hometowns to country codes.
- `run_merge_analysis.py`: joins rosters to NCAA-IPEDS crosswalk, IPEDS HD, EADA, and macro indicators.
- `fetch_extra_shocks.py`, `run_join_shocks.py`: add the eight shock panels, total population, and 15–24 cohort population.
- `run_main_results.py`: produces the extensive-margin probit and the intensive-margin count specifications reported in Tables 2 and 3.
- `run_stress_tests.py`: produces the disaster-deaths drop-top-5, refugees-on-raw-counts, and rate-model artefact comparison reported in Section 3.3.
- `run_bartik.py`, `run_event_study.py`, `run_paper_outputs.py`: build the auxiliary Bartik, COVID descriptive exercise, and Table 1 / figures.

Figures

References

Lee, D. S., J. McCrary, M. Moreira, and J. Porter (2022). “Valid t -Ratio Inference for IV.” *American Economic Review* 112(10): 3260–3290.

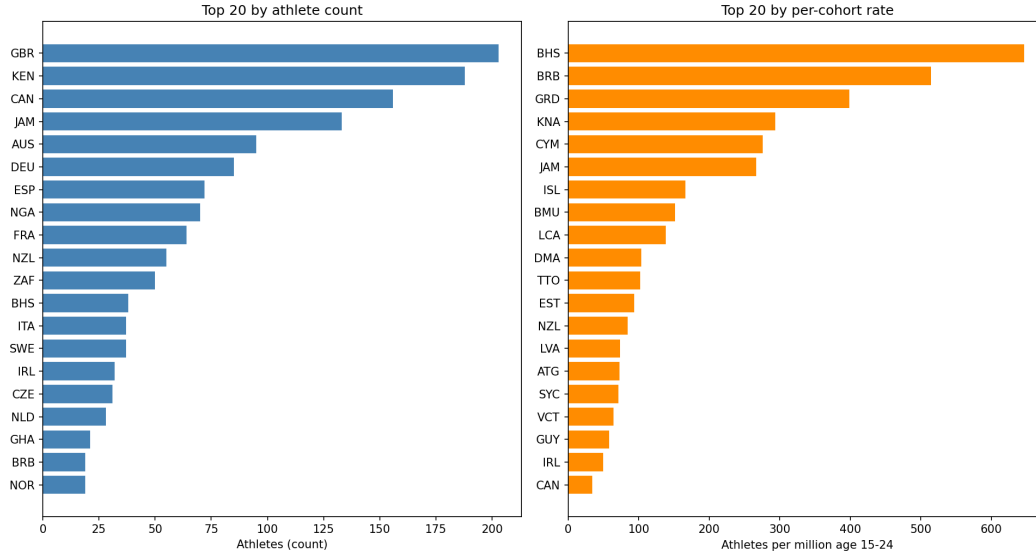


Figure 1: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right). The left panel is roughly proportional to population scale; the right panel is what the rate model implicitly maximises explanatory power on.

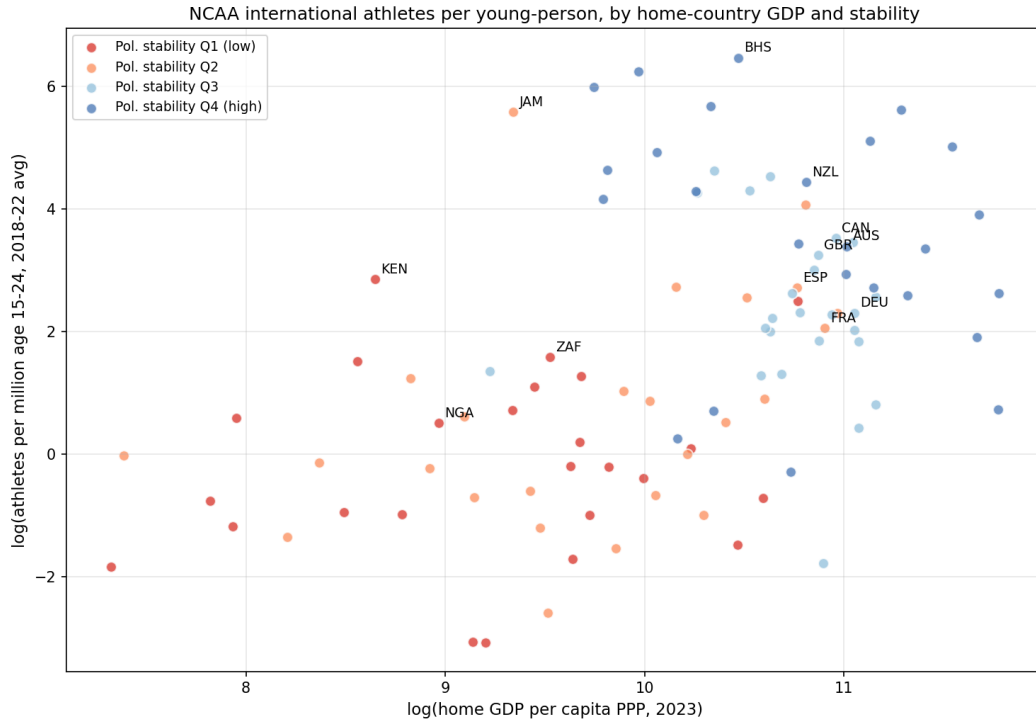


Figure 2: Log athletes per million age 15–24 vs. log home-country GDP per capita, colour-coded by political-stability quartile. Top twelve senders annotated. The apparent “political stability fans out the high-rate cluster” pattern is largely a country-size artefact: high-stability countries are systematically small, and small countries are over-represented per capita.